

Furkan Karabulut

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[Portfolio](#)

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Research Interests: Post-Quantum Cryptography, Cryptographic Engineering, Hash-Based Signatures (XMSS/LMS), Hardware-Constrained Implementations, Artificial Intelligence

EDUCATION

B.Sc. in Computer Science
North Carolina State University
GPA: 3.0/4.0

Aug/2021 – May/2024
Raleigh, NC

SKILLS

Languages:	Java, C, Python, C#, SQL, JavaScript, TypeScript, CSS, HTML
Tools:	Visual Studio, VS Code, Eclipse, JetBrains, MySQL, SSMS
Operating Systems:	Microsoft, macOS, Linux
Cryptography:	XMSS, LMS, Hash-Based Signatures, Post-Quantum Cryptography, Merkle Tree Optimization

PATENTS

- Method for Computing Unbalanced L-Trees Efficiently for Hash-Based Signatures Used in Post-Quantum Cryptographic Authentication
US20250007699A1 / WO2025005923A1 — Published Jan 2, 2025
- Method for the Implementation of Left-Node Authentication in Hash-Based Signatures XMSS and LMS
WO2025005924A1 — Published Jan 2, 2025

WORK EXPERIENCE

Associate Software Engineer, Relias LLC (Morrisville, NC)

Nov/2024 – Present

- Applied AI tooling to day-to-day engineering work, including setting up MCP servers, configuring custom AI agents and agent skills, managing context windows, and using prompt engineering techniques to integrate AI effectively into industry workflows.
- Implemented RESTful endpoints in C# and ASP.NET Core, building across controllers, handlers, entities, and DTOs against SQL Server via Entity Framework Core
- Wrote unit, integration, smoke, and BDD tests across backend services to validate endpoint functionality and maintain coverage. Documented and tested APIs using Swagger and Postman.
- Built Angular features in TypeScript, working with RxJS observables, BehaviorSubject, and subscriptions for async state management. Developed components within a micro-frontend architecture using feature flags and shared component patterns.
- Designed and deployed SQL tables and Azure Cosmos DB containers using Bicep templates. Led a technical spike to evaluate storage strategy before implementation, driving the decision to integrate CosmosDB into a backend service from scratch.
- Contributed to multi-region production deployments using GitOps tooling with Flux and Kustomize on AKS, and worked with Docker and Azure Container Registry. Wrote Azure DevOps YAML pipelines for CI/CD and coordinated releases for the team.
- Facilitated code reviews on Bitbucket, providing structured feedback on pull requests. Wrote internal documentation to help engineers ramp up on new parts of the codebase.

- Won the 2025 Relias company-wide Hackathon with an AI-powered Slack bot that streamlined team communication and task management.

Software Engineer (Intern), Live Oak Bank (Raleigh, NC)

Jun/2023 – Aug/2023

- Developed and maintained RESTful APIs focused on user authentication, tested with Postman and integrated with AWS API Gateway.
- Built dynamic, responsive frontend components in React.js, implementing UI layouts from Figma designs using CSS and Bootstrap.
- Engineered Docker containers to establish a private network for microservices communication across services.
- Researched cloud architecture and service integrations using Terraform and AWS as part of a broader infrastructure exploration initiative.
- Provided regular oral and written reports to advisors, consistently completing tasks ahead of schedule, showcasing efficiency and proactive communication within the agile framework

Software Engineer, PQSecure Technologies (Remote)

Jan/2023 – Jun/2023

- Conducted research on optimizing hash-based signature schemes XMSS and LMS for hardware-constrained environments, resulting in two sole-inventor patents.
- Received sole-inventor patent for a memory-efficient L-tree computation method in XMSS, reducing memory usage from 67 to 11 locations, a reduction applicable to hardware-constrained environments.
- Received sole inventor patent for an optimized authentication path update algorithm for XMSS and LMS, improving memory usage during signature generation
- Improved Merkle tree traversal efficiency by refining traversal logic and reducing computational overhead across signature operations.

Undergraduate Researcher, North Carolina State University (Raleigh, NC)

Jun/2022 – Nov/2022

- Achieved 96% improvement in runtime efficiency by profiling Python and Java codebases to identify time-consuming functions and rewriting them with more efficient algorithms.
- Reduced memory usage by replacing higher-cost data types (e.g., floats/doubles) with lower-footprint alternatives such as integers and fixed-size types where precision allowed
- Refactored nested loop structures and adopted more efficient data structures to improve overall performance across the codebase.

CERTIFICATES

AWS Cloud Practitioner

AZ 900